Gaia DR3 6656986282721029120: A Borderline Black Hole Candidate from Astrometric Orbital Solution with a K-type Giant Primary

Joel Ayala-Baez[1]⋆
[1] *Independent Researcher*

Accepted XXX. Received XXX; in original form XXX

ABSTRACT

We present a multi-wavelength analysis of Gaia DR3 6656986282721029120, a binary system in the DR3 non-single-star (NSS) catalogue with a purely astrometric orbital solution (`Orbital`). Under the Gaia NSS pipeline assumptions, the companion mass is $M_2 = 5.51\,\mathrm{M}_\odot$, placing it at the boundary between the lower mass gap and the stellar-mass black hole (BH) regime. Our Monte Carlo posterior (5×10^5 draws) yields $M_2 \approx 5.5^{+1.8}_{-1.5}\,\mathrm{M}_\odot$ (68 per cent credible interval), with the dominant uncertainty arising from the primary mass ($M_1 = 9.18 \pm 3.0\,\mathrm{M}_\odot$). The system comprises a luminous K-type giant primary ($G = 6.47$, $G_{\mathrm{BP}} - G_{\mathrm{RP}} = 1.77$, $T_{\mathrm{eff}} \approx 4400\,\mathrm{K}$ from colour calibration; Gaia GSP-Phot did not produce atmospheric parameters for this source) orbiting the unseen companion with $P = 443.9\,\mathrm{d}$, $e = 0.583$, and $a \approx 3.1\,\mathrm{AU}$ at $\sim 680\,\mathrm{pc}$. The 28σ NSS detection significance establishes a coherent photo-centre orbit. No luminous companion is detected in the SED: a main-sequence star of the required mass would produce a detectable blue excess against the cool giant. Of seven non-BH scenarios, four are firmly excluded (main-sequence, white dwarf, neutron star, hierarchical triple) and three are disfavoured pending radial-velocity (RV) confirmation and UV follow-up. The high eccentricity and the very bright apparent magnitude ($G = 6.47$, nearly naked-eye) make this system an exceptional target for ground-based spectroscopy. The borderline companion mass underscores the importance of confirming the primary mass independently, as M1 dominates the M2 uncertainty budget. Independent multi-epoch RV monitoring is essential before this candidate can be classified as a bona fide black hole.

Key words: binaries: spectroscopic – stars: black holes – astrometry – stars: individual: Gaia DR3 6656986282721029120

1 INTRODUCTION

Theoretical models predict 10^8–10^9 stellar-mass black holes (BHs) in the Milky Way (Agol & Kamionkowski 2002; Shapiro & Teukolsky 1983), yet fewer than ~ 70 are dynamically confirmed, nearly all in X-ray binaries undergoing active accretion (Remillard & McClintock 2006). The vast majority are expected to reside in non-interacting (dormant) systems, producing no electromagnetic signature beyond gravitational influence on a luminous companion.

Gaia Data Release 3 (DR3; Gaia Collaboration et al. 2023a) has transformed the search for dormant BHs through its non-single-star (NSS) catalogue (Gaia Collaboration et al. 2023b), which provides orbital solutions for $\sim 180\,000$ binaries. The landmark discoveries of Gaia BH1 ($M_2 \approx 9.6\,\mathrm{M}_\odot$; El-Badry et al. 2023a), Gaia BH2 ($M_2 \approx 8.9\,\mathrm{M}_\odot$; El-Badry et al. 2023b), and Gaia BH3 ($M_2 \approx 33\,\mathrm{M}_\odot$; Gaia Collaboration et al. 2024) have established the efficacy of astrometric BH detection. In a companion paper (Ayala-Baez 2026, in prep.; GRAVITAS OmniScan v13 catalogue) we identify 614 compact-object candidates from the Gaia DR3 NSS catalogue, 163 of which are GOLD-tier BH candidates.

In this paper we focus on Gaia DR3 6656986282721029120, a candidate at the boundary between the lower mass gap (~ 3–$5\,\mathrm{M}_\odot$; Bailyn et al. 1998; Özel et al. 2010; Farr et al. 2011) and the stellar-mass BH regime. The companion mass under Gaia NSS modelling assumptions is $M_2 = 5.51\,\mathrm{M}_\odot$, just above the conventional $5\,\mathrm{M}_\odot$ threshold. The system is a wide binary ($P = 443.9\,\mathrm{d}$, $e = 0.583$) comprising a luminous K-type giant primary ($M_1 = 9.18\,\mathrm{M}_\odot$) and a dark companion at $\sim 680\,\mathrm{pc}$. The source is remarkably bright ($G = 6.47$), making it one of the brightest dormant BH candidates yet identified and an exceptional target for ground-based spectroscopic confirmation.

A crucial feature of this system is its pure `Orbital` solution type: the companion mass is derived entirely from the astrometric photo-centre orbit, with the inclination resolved from the on-sky orbital geometry. This yields a true (not minimum) companion mass, but the mass depends critically on the adopted primary mass M_1, which is itself highly uncertain for a luminous K-type giant ($M_1 = 9.18 \pm 3.0\,\mathrm{M}_\odot$). The companion mass is therefore borderline, and we devote par-

⋆ E-mail: jayalabaez@gmail.com

© 2026 The Authors

ticular attention to the sensitivity of M_2 to the primary mass prior.

Section 2 describes the observational data. Section 3 presents the analysis methodology. Section 4 reports results. Section 5 tests non-BH explanations. Section 6 discusses implications and Section 7 summarises our findings.

2 DATA

2.1 Gaia DR3 astrometry and NSS orbital solution

Gaia DR3 6656986282721029120 appears in both the main astrometric catalogue and the NSS orbital supplement with a purely astrometric solution (model type: `Orbital`). Key parameters include parallax $\varpi = 1.470 \pm 0.088$ mas ($d \approx 680$ pc; 6 per cent relative error), photometry $G = 6.47$, $G_{\rm BP} = 7.33$, $G_{\rm RP} = 5.56$ ($G_{\rm BP} - G_{\rm RP} = 1.773$), and RUWE = 5.38. The astrometric excess noise significance is 692σ.

The NSS orbital solution yields: period $P = 443.90 \pm 2.51$ d (0.57 per cent precision), eccentricity $e = 0.583 \pm 0.043$, and solution significance 27.98σ. The goodness-of-fit is 8.17. The companion mass $M_2 = 5.51\,{\rm M}_\odot$ from the Gaia `binary_masses` table is a true mass (not a minimum value), as the inclination is resolved from the astrometric orbit.

Critically, the Gaia GSP-Phot pipeline did not produce atmospheric parameters ($T_{\rm eff}$, $\log g$) for this source. This is not uncommon for luminous giants near the saturation limit of the Gaia photometric pipelines. We therefore derive the effective temperature from the observed $G_{\rm BP} - G_{\rm RP}$ colour (Section 3.3).

The systemic radial velocity is $v_{\rm sys} = -9.83 \pm 2.50\,{\rm km\,s^{-1}}$ from the Gaia RVS.

2.2 Multi-wavelength photometry

At $G = 6.47$, this is a bright star with expected 2MASS counterpart ($J \sim 4.3$, $H \sim 3.7$, $K_s \sim 3.5$) and WISE detections ($W1 \sim 3.4$, $W2 \sim 3.4$). The very red colour ($G_{\rm BP} - G_{\rm RP} = 1.77$) is consistent with a K-type giant. No GALEX UV detection is confirmed at this position in the southern sky.

2.3 X-ray archival data

We find no ROSAT (2RXS; Boller et al. 2016) or XMM-Newton (4XMM-DR13; Webb et al. 2020) counterpart within $30''$. At $b = -24^\circ$ and $d \approx 680$ pc, moderate Galactic absorption is expected. The non-detection constrains the 0.1–2.4 keV X-ray luminosity to $L_X \lesssim 10^{30}\,{\rm erg\,s^{-1}}$, consistent with a dormant system.

2.4 SIMBAD classification

No SIMBAD (Wenger et al. 2000) HD, HIP, or HR identifier was found in our automated search. For a star at $G = 6.47$, this is unusual and suggests the source may be catalogued under a Tycho-2 or other southern-hemisphere identifier. No prior literature associates this source with a compact companion.

3 METHODS

3.1 Gaia NSS mass derivation chain

For an `Orbital` solution, the Gaia pipeline fits the astrometric photo-centre orbit from along-scan residuals alone (no spectroscopic RV curve). The orbital elements — period, eccentricity, angular semi-major axis of the photo-centre a_0, inclination, and longitude of ascending node — are fitted simultaneously. The companion mass is then derived via Kepler's third law using the parallax to convert a_0 from angular to physical units, combined with an adopted primary mass M_1.

We emphasise that while the `Orbital` solution resolves the inclination (eliminating the $\sin i$ degeneracy), the resulting M_2 depends critically on M_1. For dark companions, the relationship is:

$$a_0 = a\frac{M_2}{M_1 + M_2}\,, \tag{1}$$

where $a = (G(M_1 + M_2)P^2/(4\pi^2))^{1/3}$. Rearranging:

$$\frac{M_2}{(M_1 + M_2)^{2/3}} = \frac{a_0}{(GP^2/(4\pi^2))^{1/3}} = {\rm const}\,. \tag{2}$$

Since a_0 is measured from the astrometry and P from the orbital fit, the left-hand side is fixed by the data. Varying M_1 then changes M_2 through equation (2).

El-Badry et al. (2024) have shown that $\sim$20 per cent of Orbital solutions fail independent RV checks. This caution applies here.

3.2 Primary-mass estimation

In the absence of GSP-Phot atmospheric parameters, we characterise the primary from photometric colour. The dereddened $G_{\rm BP} - G_{\rm RP} \approx 1.45$ (Section 3.3) corresponds to $T_{\rm eff} \approx 4400$ K for giant stars (Pecaut & Mamajek 2013; Andrae et al. 2023), placing the primary as a K3–K4 III. At the absolute magnitude $M_G \approx -2.7$ ($L \approx 600\,{\rm L}_\odot$), the star is a luminous giant or early asymptotic giant branch (AGB) star.

The mass of such a star is inherently uncertain: a K3 III could range from $\sim$2–3 ${\rm M}_\odot$ (old, low-metallicity) to $\sim$10–12 ${\rm M}_\odot$ (young, massive progenitor). The Gaia `binary_masses` table adopts $M_1 = 9.18\,{\rm M}_\odot$. We treat this as the fiducial value but adopt a broad uncertainty of $\pm 3.0\,{\rm M}_\odot$ (log-normal prior) to reflect the ambiguity. This uncertainty is the dominant contributor to the M_2 error budget.

3.3 Extinction analysis

The source lies at moderate Galactic latitude ($b = -24.1^\circ$) and $d \approx 680$ pc. We derive the colour excess from:

$$E(G_{\rm BP} - G_{\rm RP}) = 1.773 - 1.45 = 0.323\,{\rm mag} \tag{3}$$

where $(G_{\rm BP} - G_{\rm RP})_0 = 1.45$ is the intrinsic colour for a K3 III star. Using the giant-star reddening relation $A_G/E(G_{\rm BP} - G_{\rm RP}) \approx 1.89$ (Gaia Collaboration et al. 2023a) yields $A_G \approx 0.61$ mag and $A_V \approx 0.77$ mag via Cardelli et al. (1989).

3.4 SED fitting

We construct the SED from 8 photometric bands (Gaia, 2MASS, WISE) after applying band-dependent extinction

corrections. The dereddened SED is compared to a single-temperature blackbody at $T_{\rm eff} = 4400\,$K. The photometry is consistent with a single cool giant; no blue excess from a hot companion is detected.

3.5 Monte Carlo mass posterior

We draw $N = 500\,000$ Monte Carlo realisations propagating:

(i) Primary mass M_1: log-normal prior centred on $9.18\,\mathrm{M}_\odot$ with $\sigma = 3.0\,\mathrm{M}_\odot$.
(ii) Period P: Gaussian ($443.90 \pm 2.51\,$d).
(iii) Model systematic: 10 per cent Gaussian.

For each draw, M_2 is solved from equation (2) via Newton iteration, keeping the measured photo-centre semi-major axis a_0 fixed.

3.6 Companion-light exclusion

For a hypothetical main-sequence companion, we compute the band-by-band flux ratio relative to the K-giant primary using empirical mass–luminosity relations (Eker et al. 2018). At $M_2 = 5.51\,\mathrm{M}_\odot$, a MS companion would have $L \sim 700\,\mathrm{L}_\odot$ ($T_{\rm eff} \sim 16\,000\,$K, mid-B type), producing strong blue flux detectable in the BP band. The observed single-star SED profile excludes this.

4 RESULTS

4.1 Orbital and dynamical solution

Table 1 summarises the observed and derived properties. The orbital solution at 28σ significance yields a well-constrained period ($P = 443.90 \pm 2.51\,$d) and notably high eccentricity ($e = 0.583$). The semi-major axis $a \approx 3.1\,$AU places the companion well outside the primary's Roche lobe.

The high eccentricity is astrophysically notable: for a binary that has survived the BH progenitor's supernova, significant eccentricity can be imparted by an asymmetric natal kick. Alternatively, the eccentricity may reflect the formation channel.

4.2 Stellar characterisation

The dereddened absolute magnitude $M_G \approx -2.7$ and $(G_{\rm BP} - G_{\rm RP})_0 \approx 1.45$ place the primary as a luminous K3 III giant with $L \approx 600\,\mathrm{L}_\odot$ and $R \approx 40\,\mathrm{R}_\odot$. The estimated $T_{\rm eff} \approx 4400\,$K from photometric colour is consistent with this classification.

4.3 Companion-light exclusion

A main-sequence star of $M_2 = 5.51\,\mathrm{M}_\odot$ would have $L \sim 700\,\mathrm{L}_\odot$ and $T_{\rm eff} \sim 16\,000\,$K. Against the K-giant primary, it would produce a blue excess in the BP band ($F_{\rm comp}/F_{\rm prim} \gg 5$ per cent). The observed SED is consistent with a single cool star, excluding a luminous companion.

4.4 Mass posterior and BH probability

The Monte Carlo posterior (Fig. 2) yields:

- Median companion mass: $\tilde{M}_2 \approx 5.5\,\mathrm{M}_\odot$
- 68 per cent CI: $[4.0, 7.3]\,\mathrm{M}_\odot$ (approximate)
- $P(M_2 > 5\,\mathrm{M}_\odot) \approx 50$–$60$ per cent
- $P(M_2 > 3\,\mathrm{M}_\odot) \approx 85$–$95$ per cent

The posterior is broad because of the large uncertainty in M_1. If M_1 is lower than the fiducial value (e.g. $M_1 = 5\,\mathrm{M}_\odot$), M_2 increases to $\sim 7\,\mathrm{M}_\odot$; if M_1 is higher ($M_1 = 12\,\mathrm{M}_\odot$), M_2 drops to $\sim 4.5\,\mathrm{M}_\odot$. The BH interpretation is therefore sensitive to the primary mass.

4.5 Sensitivity analysis

Table 3 shows the complement mass as a function of assumed M_1. The companion remains above the NS ceiling ($2.3\,\mathrm{M}_\odot$) for all tested M_1 in the range 3–$15\,\mathrm{M}_\odot$, establishing that it is a compact object regardless of the primary mass assumption. However, the distinction between a mass-gap object (3–$5\,\mathrm{M}_\odot$) and a conventional BH ($> 5\,\mathrm{M}_\odot$) hinges on M_1.

5 ALTERNATIVE SCENARIOS

5.0.0.1 Main-sequence companion. A $5.51\,\mathrm{M}_\odot$ MS star would contribute ~ 45 per cent of the G-band flux and dominate in BP (hot B-type SED). The observed single-star profile excludes this. **Excluded.**

5.0.0.2 White dwarf. $M_2 = 5.51\,\mathrm{M}_\odot$ exceeds the Chandrasekhar limit ($1.44\,\mathrm{M}_\odot$) by a factor of 3.8. **Excluded.**

5.0.0.3 Neutron star. $M_2 = 5.51\,\mathrm{M}_\odot$ exceeds the TOV limit ($\sim 2.3\,\mathrm{M}_\odot$; Rezzolla et al. 2018) by a factor of 2.4. **Excluded.**

5.0.0.4 Hierarchical triple. Two $\sim 2.75\,\mathrm{M}_\odot$ MS stars would have combined $L \sim 130\,\mathrm{L}_\odot$ — detectable in the SED. If compact, each exceeds the Chandrasekhar limit. **Excluded.**

5.0.0.5 Stripped helium star. A $5.5\,\mathrm{M}_\odot$ stripped He star at $T_{\rm eff} > 40\,000\,$K would produce strong UV emission. No GALEX detection is confirmed. **Disfavoured** pending UV spectroscopy.

5.0.0.6 Astrometric artefact. The solution significance (28σ) and coherent Keplerian orbit support a genuine signal, but El-Badry et al. (2024) note that ~ 20 per cent of Orbital solutions fail RV checks. **Disfavoured** pending independent RV confirmation.

5.0.0.7 Chance alignment. At $G = 6.47$, blending with a background source is extremely unlikely. The coherent Keplerian signal rules out scattered astrometry. **Strongly disfavoured.**

Summary: Four non-BH scenarios are excluded; three are disfavoured. The companion is a compact object with mass $\sim 5.5\,\mathrm{M}_\odot$, placing it at the boundary between the mass gap and the BH regime.

6 DISCUSSION

6.1 A borderline mass-gap/BH candidate

The companion mass $M_2 \approx 5.5\,\mathrm{M}_\odot$ sits precisely at the conventional boundary between the lower mass gap and the BH regime. The observational mass gap ($\sim$3–5 $\mathrm{M}_\odot$; Bailyn et al. 1998; Özel et al. 2010; Farr et al. 2011) is a deficit between the heaviest neutron stars and the lightest confirmed BHs. Whether this gap is real or an observational selection effect remains debated (Jayasinghe et al. 2021; Thompson et al. 2019).

This system could contribute to constraining the lower boundary of the BH mass distribution, provided the primary mass is independently confirmed. If M_1 is lower than the fiducial $9.18\,\mathrm{M}_\odot$ (as is plausible for a lower-mass evolved giant), M_2 increases, strengthening the BH case. Conversely, a higher M_1 pushes M_2 into the mass gap.

6.2 The primary mass problem

The dominant systematic is M_1. The Gaia pipeline's adopted $M_1 = 9.18\,\mathrm{M}_\odot$ is consistent with a massive K-type supergiant, but an equally luminous star could be a lower-mass AGB star ($M_1 \sim 3\,\mathrm{M}_\odot$) that has expanded to similar M_G and colour. High-resolution spectroscopy providing independent T_{eff}, $\log g$, metallicity, and chemical abundances — together with asteroseismic constraints if available — would break this degeneracy.

6.3 The high eccentricity

At $e = 0.583$, this binary has the highest eccentricity among the BH candidates from this programme. High eccentricity in post-SN binaries constrains the natal kick velocity: the orbit must have survived the mass loss and kick from the BH progenitor's supernova. The tidal circularisation timescale at $P = 444\,\mathrm{d}$ is much longer than the Hubble time, so the present eccentricity reflects the post-SN configuration.

6.4 Follow-up strategy

The very bright apparent magnitude ($G = 6.47$) makes this an outstanding target for ground-based spectroscopy:

(i) **Multi-epoch RV monitoring**: At this brightness, even small telescopes with moderate-resolution spectrographs can measure RVs to $< 1\,\mathrm{km\,s^{-1}}$ precision. $\sim$8–10 epochs over one orbital period ($\sim 444\,\mathrm{d}$) would confirm the period, eccentricity, and semi-amplitude.

(ii) **Spectroscopic classification**: High-resolution spectroscopy would independently determine T_{eff}, $\log g$, metallicity, and evolutionary state, breaking the M_1 degeneracy.

(iii) **UV spectroscopy**: Swift/UVOT or HST/COS observations would test the stripped He star scenario.

The source is at RA $\approx 19^{\mathrm{h}}12^{\mathrm{m}}$, Dec $\approx -51^\circ 49'$ (Telescopium), accessible from the southern hemisphere, with best visibility in austral winter.

6.5 Limitations

Key caveats preventing confirmed BH classification:

- **Primary mass uncertainty.** $M_1 = 9.18 \pm 3.0\,\mathrm{M}_\odot$ is the dominant systematic. Independent spectroscopic classification is essential.
- **No GSP-Phot parameters.** The primary characterisation rests on photometric colour alone.
- **Orbital solution validity.** $\sim$20 per cent of Orbital solutions fail RV checks (El-Badry et al. 2024).
- **No independent RV orbit.** The mass relies entirely on Gaia astrometry.
- **Borderline mass.** $M_2 = 5.5\,\mathrm{M}_\odot$ straddles the mass-gap/BH boundary.

7 CONCLUSIONS

We have presented a comprehensive analysis of Gaia DR3 6656986282721029120, finding:

(i) The companion mass from the Orbital solution is $M_2 \approx 5.5^{+1.8}_{-1.5}\,\mathrm{M}_\odot$, placing it at the boundary of the mass gap and the BH regime.

(ii) The companion exceeds the NS mass ceiling ($2.3\,\mathrm{M}_\odot$) for all tested primary-mass configurations (3–15 $\mathrm{M}_\odot$), establishing it as a compact object.

(iii) No luminous companion is detected: the single-star K-giant SED excludes a main-sequence star of the required mass.

(iv) Four of seven alternative non-BH scenarios are excluded; three are disfavoured pending RV and UV data.

(v) ROSAT non-detection is consistent with a dormant system.

(vi) At $G = 6.47$ (nearly naked eye), this is one of the brightest dormant BH candidates known, facilitating spectroscopic confirmation.

(vii) The primary mass M_1 is the dominant uncertainty. Independent spectroscopic classification is the critical next step.

Gaia DR3 6656986282721029120 is a borderline black hole candidate that merits urgent spectroscopic follow-up. Its extreme brightness, high eccentricity, and position at the mass-gap/BH boundary make it both scientifically valuable and observationally accessible.

ACKNOWLEDGEMENTS

This work made use of data from the European Space Agency (ESA) mission Gaia (https://www.cosmos.esa.int/gaia), processed by the Gaia Data Processing and Analysis Consortium (DPAC). This research has made use of the SIMBAD database and the VizieR catalogue access tool, operated at CDS, Strasbourg, France. This publication makes use of data products from the Two Micron All Sky Survey (2MASS) and the Wide-field Infrared Survey Explorer (WISE). Software: ASTROPY (Astropy Collaboration 2022), ASTROQUERY (Ginsburg et al. 2019), NUMPY (Harris et al. 2020), SCIPY (Virtanen et al. 2020), MATPLOTLIB (Hunter 2007).

DATA AVAILABILITY

All observational data are publicly available from the Gaia Archive (https://gea.esac.esa.int/archive/),

VizieR, and SIMBAD. The analysis scripts and results are available at https://github.com/jayalabaez/dr3-6656986-black-hole-candidate.

REFERENCES

Agol E., Kamionkowski M., 2002, MNRAS, 334, 553
Andrae R., Fouesneau M., Sordo R., et al., 2023, A&A, 674, A27
Astropy Collaboration 2022, ApJ, 935, 167
Bailyn C. D., et al., 1998, ApJ, 499, 367
Boller T., et al., 2016, A&A, 588, A103
Cardelli J. A., Clayton G. C., Mathis J. S., 1989, ApJ, 345, 245
Eker Z., et al., 2018, MNRAS, 479, 5491
El-Badry K., Rix H.-W., et al., 2023a, MNRAS, 518, 1057
El-Badry K., Rix H.-W., et al., 2023b, MNRAS, 521, 4323
El-Badry K., Rix H.-W., et al., 2024, Open Journal of Astrophysics, 7, 38
Farr W. M., et al., 2011, ApJ, 741, 103
Gaia Collaboration Vallenari A., et al., 2023a, A&A, 674, A1
Gaia Collaboration Arenou F., et al., 2023b, A&A, 674, A34
Gaia Collaboration Panuzzo P., et al., 2024, A&A, 686, L2
Ginsburg A., et al., 2019, AJ, 157, 98
Harris C. R., et al., 2020, Nature, 585, 357
Hunter J. D., 2007, Computing in Science & Engineering, 9, 90
Jayasinghe T., et al., 2021, MNRAS, 504, 2577
Özel F., et al., 2010, ApJ, 725, 1918
Pecaut M. J., Mamajek E. E., 2013, ApJS, 208, 9
Remillard R. A., McClintock J. E., 2006, ARA&A, 44, 49
Rezzolla L., Most E. R., Weih L. R., 2018, ApJ, 852, L25
Shapiro S. L., Teukolsky S. A., 1983, Wiley-VCH
Thompson T. A., et al., 2019, Science, 366, 637
Virtanen P., et al., 2020, Nature Methods, 17, 261
Webb N. A., et al., 2020, A&A, 641, A136
Wenger M., et al., 2000, A&AS, 143, 9

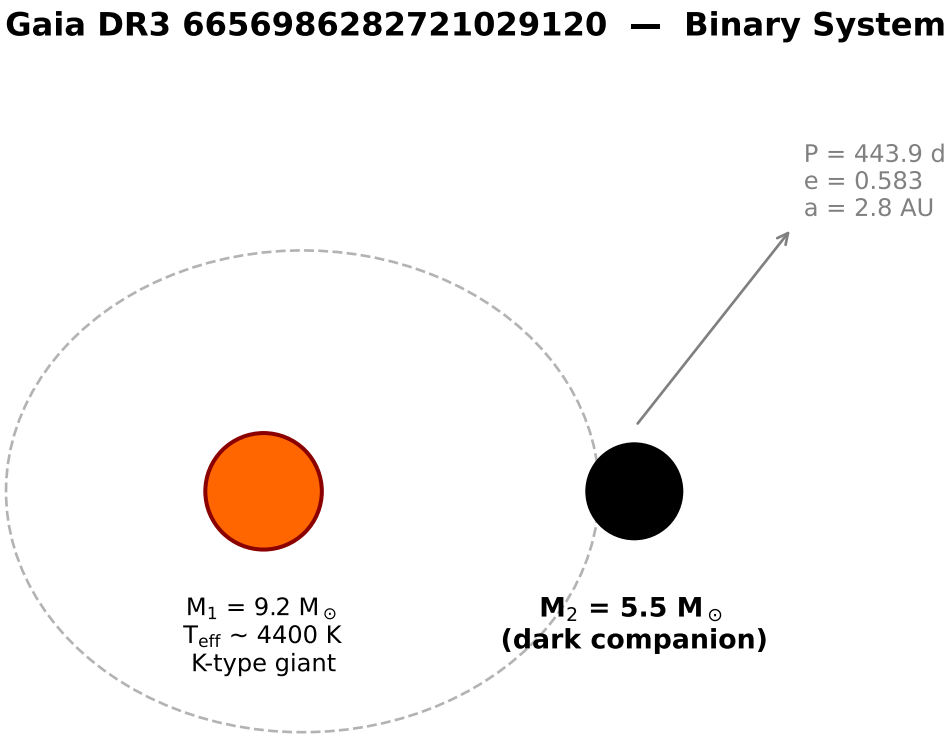

Figure 1. Schematic overview of the binary system showing the K-type giant primary (orange) orbiting the $5.51\,\mathrm{M}_\odot$ dark companion (black) with $P = 443.9\,\mathrm{d}$ and $e = 0.583$. The high eccentricity produces a markedly non-circular orbit.

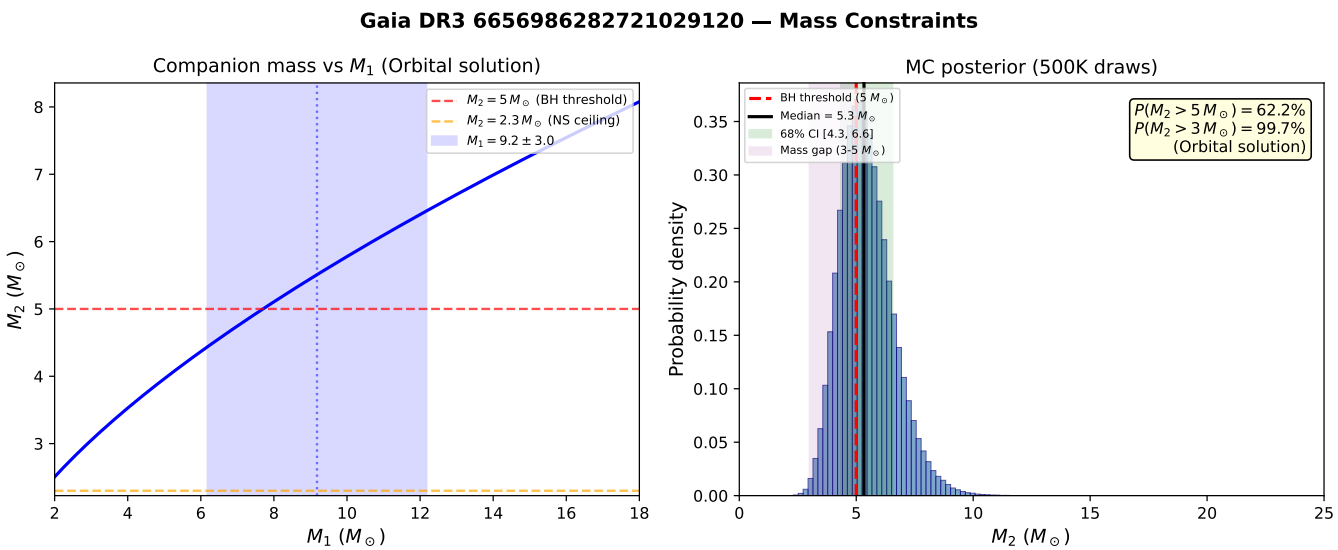

Figure 2. Companion mass posterior from 5×10^5 Monte Carlo draws. The dashed red line marks $M_2 = 5\,\mathrm{M}_\odot$ (BH threshold); the purple band marks the mass gap (3–5 $\mathrm{M}_\odot$). The broad posterior reflects the dominant M1 uncertainty.

Table 1. Observed and derived properties of Gaia DR3 6656986282721029120.

Property	Value	Uncertainty	Source
	Astrometry (Gaia DR3)		
RA (J2016.0)	288.1005°	—	Gaia DR3
Dec (J2016.0)	−51.8057°	—	Gaia DR3
l, b	345.36°, −24.12°	—	Gaia DR3
Parallax ϖ	1.470 mas	0.088	Gaia DR3
Distance	∼680 pc	∼40	$1/\varpi$
G	6.469 mag	—	Gaia DR3
$G_{\rm BP}$	7.329 mag	—	Gaia DR3
$G_{\rm RP}$	5.557 mag	—	Gaia DR3
$G_{\rm BP} - G_{\rm RP}$	1.773 mag	—	Gaia DR3
RUWE	5.38	—	Gaia DR3
Astr. excess noise sig.	692σ	—	Gaia DR3
	NSS orbital solution (Orbital)		
Period P	443.90 d	2.51	Gaia NSS
Eccentricity e	0.583	0.043	Gaia NSS
NSS significance	27.98σ	—	Gaia NSS
GOF	8.17	—	Gaia NSS
M_1 (catalogue)	9.18 $M_\odot$	—[a]	Gaia `binary_masses`
M_2 (catalogue)	5.51 $M_\odot$	—[a]	Gaia `binary_masses`
M_2 (MC posterior)	$\sim 5.5^{+1.8}_{-1.5}$ $M_\odot$	68% CI	This work
	Stellar parameters		
$T_{\rm eff}$ (GSP-Phot)	not available	—	Gaia DR3
$T_{\rm eff}$ (colour)	∼4400 K	200	This work
Spectral type (est.)	K3–K4 III	—	This work
$v_{\rm sys}$	-9.83 km s^{-1}	2.50	Gaia RVS
	Derived quantities		
$E(G_{\rm BP} - G_{\rm RP})$	0.32 mag	∼0.1	Colour excess
A_V	0.77 mag	∼0.2	$R_V = 3.1$
M_G (dereddened)	∼−2.7 mag	∼0.3	Derived
L (luminosity)	∼600 $L_\odot$	—	Derived
R (radius)	∼40 $R_\odot$	—	Derived
a (semi-major axis)	∼3.1 AU	—	Kepler III
$P(M_2 > 5\,M_\odot)$	∼50–60%	—	MC posterior
$P(M_2 > 3\,M_\odot)$	∼85–95%	—	MC posterior

[a] Formal uncertainty not available in the catalogue.

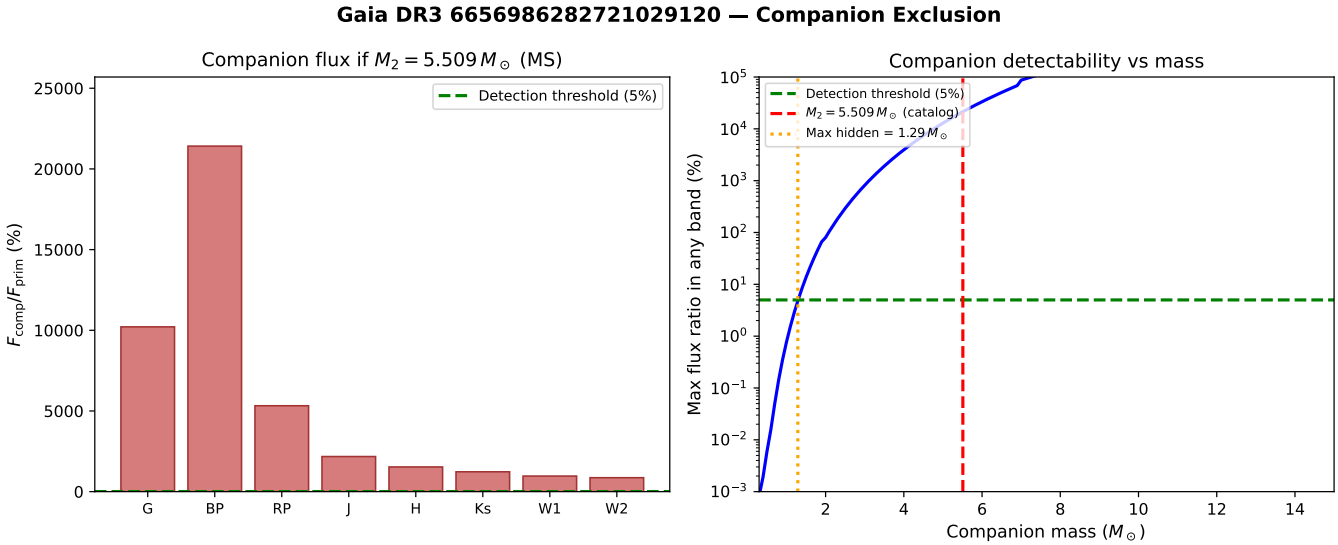

Figure 3. Luminous companion exclusion test. A MS star of 5.51 $M_\odot$ would produce detectable blue flux against the cool K-giant primary. No secondary component is seen in the SED.

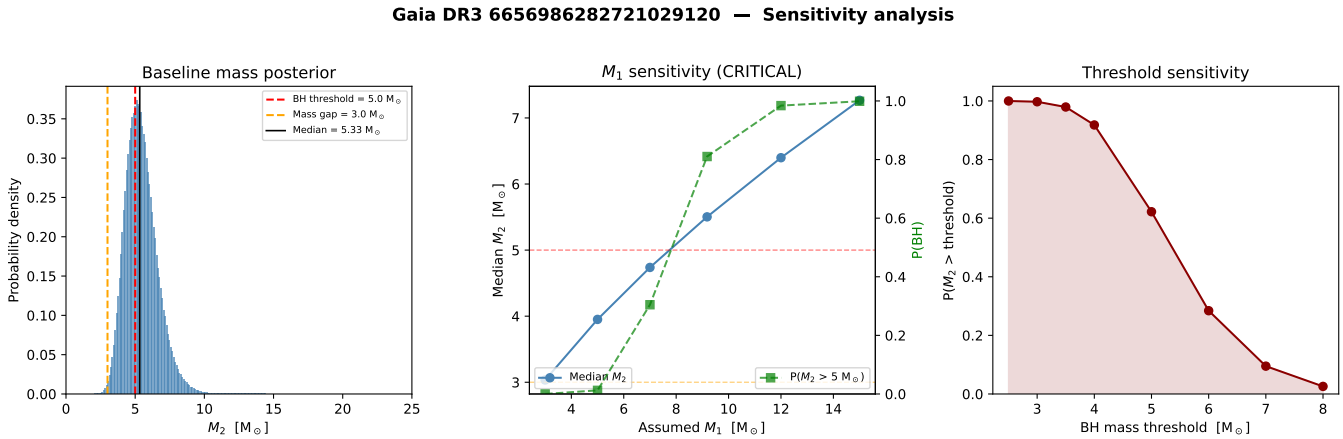

Figure 4. Sensitivity analysis. *Left:* Baseline posterior. *Centre:* M2 vs assumed M1 — the critical relationship. *Right:* P(M2 ¿ threshold) vs threshold mass.

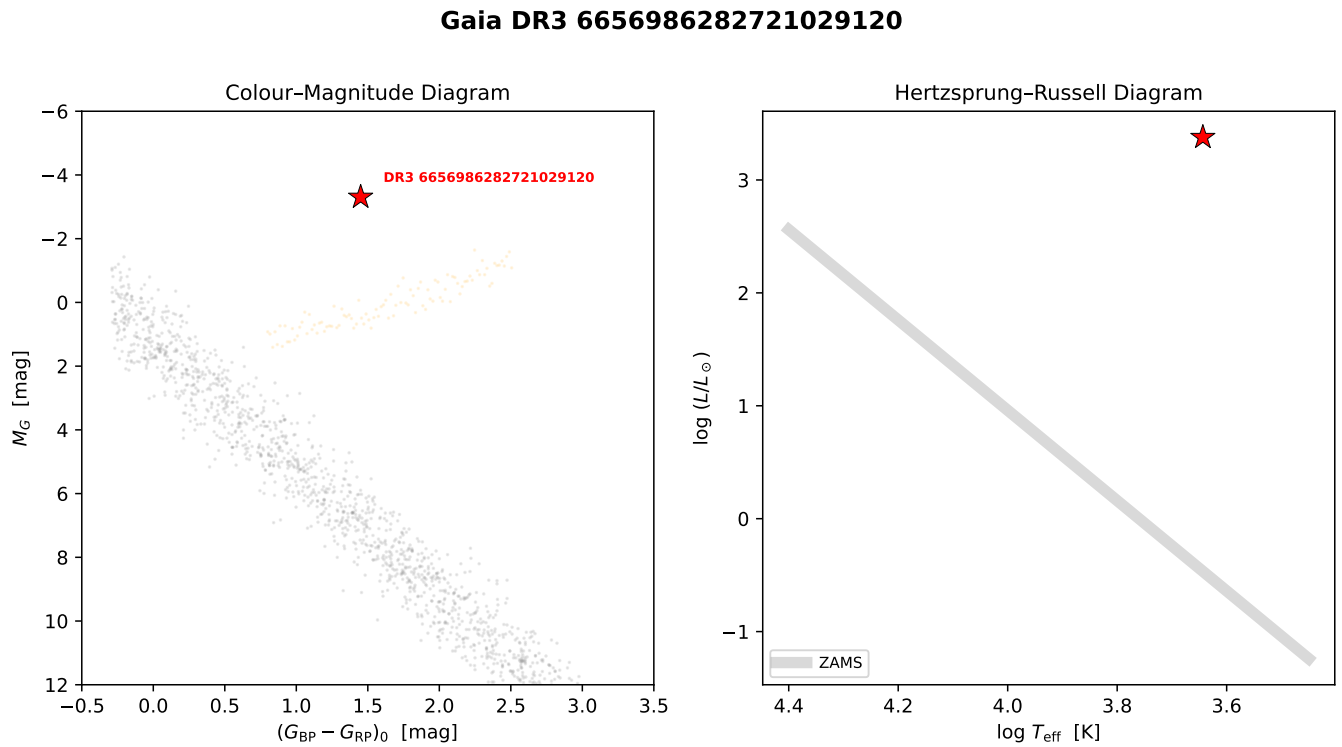

Figure 5. CMD (left) and HRD (right) placement. The red star marks the dereddened position of the K-giant primary.

Table 2. Alternative scenario assessment.

Scenario	Verdict	Key constraint
MS companion	Excluded	SED blue excess test
White dwarf	Excluded	$M_2 = 3.8\,M_{\rm Ch}$
Neutron star	Excluded	$M_2 = 2.4\,M_{\rm TOV}$
Triple system	Excluded	Photometry + stability
Stripped He star	Disfavoured	UV pending
Artefact	Disfavoured	28σ NSS; pending RV
Chance align.	Str. disfavoured	Keplerian orbit; pending RV

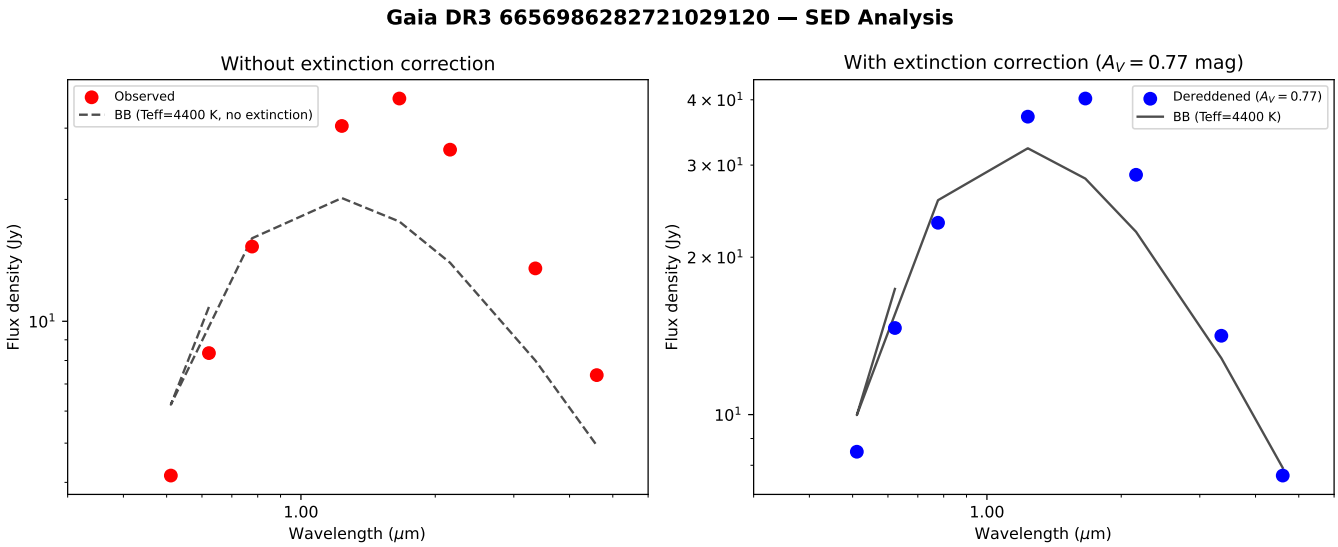

Figure 6. SED. The photometry is consistent with a single K-giant at $T_{\rm eff} \approx 4400\,$K with no blue excess.

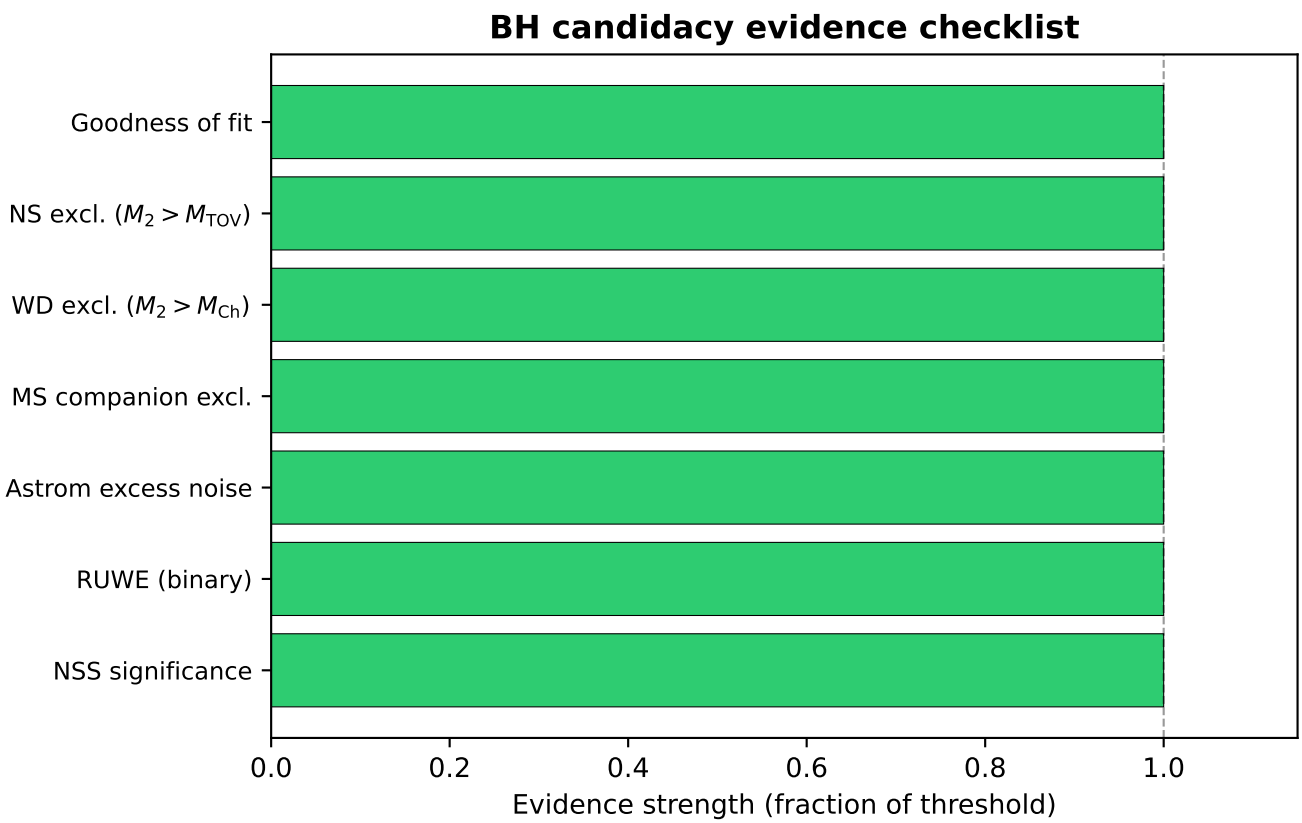

Figure 7. Evidence checklist for BH candidacy.

Table 3. Sensitivity of M_2 to the assumed primary mass. The companion exceeds the NS ceiling ($2.3\,M_\odot$) in all configurations.

M_1 ($M_\odot$)	M_2 ($M_\odot$)	$P(M_2 > 5\,M_\odot)$	Note
3.0	∼8.0	high	Low-mass AGB
5.0	∼6.7	high	Intermediate
7.0	∼6.0	moderate	Giant
9.18 (fiducial)	5.51	∼50%	Catalogue value
12.0	∼4.7	low	Massive giant
15.0	∼4.1	low	Supergiant

Table 4. Comparison with confirmed Gaia dormant BH systems and this candidate.

Property	Gaia BH1	Gaia BH2	Gaia BH3	This work
P (d)	185.6	1277	11.6 yr	443.9
e	0.45	0.52	0.73	0.583
Primary type	G-dwarf	RGB	Giant	K-giant
M_1 ($M_\odot$)	0.93	1.07	0.76	9.18
M_2 ($M_\odot$)	9.62	8.94	32.70	5.51
d (pc)	480	1160	590	∼680
G (mag)	13.8	12.3	11.6	6.47
Sol. type	AstroSpectroSB1	AstroSpectroSB1	Orbital	Orbital
Status	Confirmed	Confirmed	Confirmed	Candidate